O'REILLY

Chapter 13 Responsible Artificial Intelligence Applications for Social Good

Ozlem Garibay[1] *and Brent Winslow*[2]

[1] *Industrial Engineering and Management Systems, University of Central Florida, Orlando, FL, USA*

[2] *Google, Mountain View, CA, USA*

13.1 Introduction

Artificial intelligence (AI) has rapidly evolved from a futuristic concept to an integral part of contemporary life, influencing how we interact with each other and the environment (West and Allen 2018). Amidst this technological surge, responsible AI has emerged, emphasizing the creation and utilization of AI systems that uphold human rights, ethical principles,

and moral values and prioritize human and environmental well-being and safety (Artificial Intelligence Index 2024). Concurrently, harnessing the power of AI for social good has gained traction, focusing on deploying AI technologies that advance human rights, promote well-being and safety, improve environmental conditions, foster development, and enhance living conditions globally (West and Allen 2018).

As enshrined in the Universal Declaration of Human Rights (United Nations General Assembly 1949), human rights are a broad range of fundamental rights and freedoms to which all human beings are entitled. These rights are generally categorized into two main groups: (i) civil and political rights: fundamental rights such as the right to life, freedom from slavery, and privacy. These rights are typically more straightforward to attain and enforce due to their more clearly defined nature. (ii) Economic, social, and cultural (ESC) rights: these encompass rights such as access to medical care, education, housing, and an adequate standard of living. Achieving ESC rights presents greater challenges, often due to the lack of universally agreed-upon standards and the extensive resources required for their fulfillment (OHCHR 2024).

The United Nations Sustainable Development Goals (SDGs) serve as a framework for advancing ESC rights, offering a general guideline for human development that aligns with these rights (SDGs 2016). By adopting the SDGs, nations and organizations can systematically work toward sustainable and equitable development, leveraging AI as a powerful tool in this endeavor (United States Department of State 2023).

This chapter describes the principles, opportunities, and challenges of leveraging AI to improve society and the environment. AI for social good refers to the strategic application of AI technologies to address and solve pressing societal challenges (Tomašev et al. 2020). This approach involves harnessing AI's potential to promote human rights, foster development, and improve living conditions across various domains such as healthcare, education, environmental sustainability, and humanitarian efforts.

For instance, AI is revolutionizing healthcare by enhancing diagnostic accuracy, personalizing treatment plans, and predicting disease outbreaks. Increasingly, AI-powered imaging tools can detect diseases like cancer at

early stages, significantly improving patient outcomes (Tobore 2019). Moreover, AI-driven predictive analytics can forecast epidemics, enabling timely interventions and resource allocation (Dogheim and Hussain 2023). AI plays a pivotal role in environmental conservation and sustainability efforts. It aids in monitoring and analyzing environmental data, optimizing resource usage, and mitigating the impacts of climate change. For example, AI can predict and manage natural disasters by analyzing weather patterns and seismic data, reducing casualties and economic losses. In the realm of education, AI facilitates personalized learning experiences, bridging educational gaps and promoting inclusive learning. AI-powered platforms can adapt to individual learning styles and paces, providing customized content and real-time feedback. Additionally, AI can support teachers by automating administrative tasks, providing them more time to focus on student engagement and instruction (Milberg 2024). AI enhances humanitarian efforts by providing better disaster prediction, monitoring, and response. Increasingly, AI is allowing more diverse data from disparate sources to combine in new ways to support humanitarian efforts and disaster response.

The development of responsible AI for social good is a collective endeavor, requiring the collaborative efforts of multiple stakeholder groups, including researchers, developers, business leaders, and policymakers. Each group plays a critical role in ensuring that AI technologies are designed and implemented ethically and effectively. Addressing major challenges of developing AI for good, including ensuring that applications are focused on human well-being, designed responsibly, privacy and governance, enabled with robust evaluations to support humanity and the environment, also requires the collective efforts of all stakeholders (Ozmen Garibay et al. 2023). By working together, these stakeholder groups can create a comprehensive ecosystem that supports the development of responsible AI, ultimately leading to AI systems that advance human rights, promote ethical standards, and contribute to the overall well-being of society (Figure 13.1).



Figure 13.1 AI for supporting human and environmental well-being and safety.

Source: Adapted from Ozmen Garibay et al. (2023). United Nations Sustainable Development Goals (https://www.un.org/sustainabledevelopment). The content of this publication has not been approved by the United Nations and does not reflect the views of the United Nations or its officials or Member States.

13.2 Ethical Aspects of AI for Social Good Applications

The deployment of AI for social good has the potential to drive transformative change across various sectors by addressing complex societal challenges. However, the development and application of AI technologies raise critical ethical considerations that need to be addressed to ensure these advancements truly benefit society.

13.2.1 Definitions, Applications, and Mitigation Strategies for Bias and Fairness

Fairness refers to the absence of bias toward an individual or a group based on their inherent or acquired attributes that are irrelevant to the outcomes of a task (Verma and Rubin, 2018; Makhlouf et al. 2021; Mehrabi et al. 2021; Mitchell et al. 2021; Castelnovo et al. 2022). Bias occurs when AI systems produce results that disproportionately benefit certain groups or individuals over others. Concerns related to fairness in AI systems, in particular machine learning (ML) algorithms, have been growing as the applications of AI broaden. Applications of fairness in AI are widespread, including in finance for loan approval, healthcare for diagnosing diseases, criminal justice for recidivism prediction, and many others. As AI becomes more integral to decision-making processes, ensuring fairness is crucial to prevent systematic inequality and injustice. Studies addressing the concerns in fairness in AI and ML focus on how to measure fairness, identify and mitigate problems. A growing number of definitions in the literature underline the diverse and complex nature of the fairness concept and its dependence on its context. Case studies in fairness in AI, spanning various domains, reveal the complex nature of biases. One prominent example is the use of AI in hiring, where algorithms may inadvertently favor candidates based on gender or ethnicity due to biased training data (Bogen 2019; Chen 2023). Another is facial recognition technology, which has been shown to have lower accuracy rates for certain demographic groups, raising concerns about equitable treatment in surveillance and law enforcement (Najibi 2020; Chin-Rothmann and Lee 2022; Blatt 2024). Similarly, financial institutions have encountered issues with AI algorithms used in loan decisions, which unfairly disadvantaged certain demographic groups (Andrews 2021; Hale 2021; Garcia et al. 2023).

Such instances underscore the critical need for fairness, ensuring that AI-based decisions do not perpetuate unjust disparities among diverse demographic cohorts. Mitigation strategies for fairness in AI involve a combination of technical and regulatory measures. Technical strategies include preprocessing approaches such as de-biasing training data (Rajabi and Garibay 2021, 2022; Mutlu et al. 2022; Tayebi et al. 2022), in-process-

ing approaches such as developing fairness-aware algorithms (Nam et al. 2020; Ramaswamy et al. 2021; Pessach and Shmueli 2022; Wan et al. 2023; Caton and Haas 2024), and postprocessing approaches such as continuous monitoring for bias in deployed AI systems. Regulatory strategies include creating and enforcing policies that promote transparency, accountability, and inclusive practices in AI development and usage. Maintaining transparency throughout the AI lifecycle, from development to deployment, empowers stakeholders to identify and rectify biases. Additionally, engaging diverse teams in AI development and fostering interdisciplinary dialogue on ethical AI can contribute to more equitable AI systems (CSET 2021; GOV.UK 2021; METI 2022; The White House 2022; Innovation, Science and Economic Development Canada 2023; Infocomm Media Development Authority 2024; EU Artificial Intelligence Act 2024).

13.2.2 Definitions, Applications, and Mitigation Strategies for Interpretability and Explainability

Trust in AI is paramount for social good initiatives, as it forms the foundation of public confidence in technologies designed to benefit society. Trustworthiness in AI is closely linked with interpretability and explainability. For instance, in a healthcare setting, an AI model predicting patient risks must be interpretable, and its decision process must be explainable to clinicians (Holzinger et al. 2017). Interpretability refers to the ease with which humans can understand a model's internal workings and its input-output relationships. Models, such as linear regression and decision trees, are typically considered interpretable due to their straightforward nature. However, the degree of interpretability can vary depending on the complexity of the model, such as the number of features it incorporates. Essentially, interpretable AI systems enable users to comprehend the mechanisms behind their decisions, facilitating the validation of these systems against established ethical and operational standards. Explainability, on the other hand, pertains to understanding models through external mechanisms and observations. Both interpretability and explainability enhance trust by making the rationale behind AI decisions transparent, ensuring that outcomes are aligned with user intentions and societal norms, thereby instilling a sense of trust and confidence in the audience, particularly the healthcare professionals.

Developing mitigation strategies and methodologies to improve the explainability of AI algorithms has been an active area of interest for research. Various techniques have been developed. Partial dependence plots (Friedman 2001; Goldstein et al. 2015; Holzinger et al. 2017; Greenwell et al. 2018). Accumulated local effects (ALE) provide a visual representation of one input feature on the outcome (Apley and Zhu 2020). Local interpretable model-agnostic explanations analyze interactions between features by performing multifeature perturbations for a particular output (Ribeiro et al. 2016). SHapley Additive exPlanations (SHAP) leverage the game-theoretical approach to assign Shapley values for features based on their impact on the output (Lundberg and Lee 2017). Deep learning (DL) important features uses backpropagation to identify neurons with higher contributions to the output. The contrastive explanations method provides constructive explanations regarding what inputs should be minimally and sufficiently present to justify the output (Dhurandhar et al. 2018).

There are a growing number of toolkits that have been made available that include these approaches, such as AI Explainability 360 (AIX360) (AI Explainability 360, 2023), InterpretML (*InterpretML, 2023*), Alibi Explain (Bhatt et al. 2020), What-if Tool (WIT) (*What-If Tool*, 2024), and Model Agnostic Language for Exploration and Explanation (DALEX) (Biecek 2018).

To enhance interpretability and explainability, developers can incorporate model-agnostic explanation techniques and user-friendly interfaces that allow stakeholders to question and understand AI decisions (Guidotti et al. 2018). Engaging domain experts during the development process can also ensure that explanations are meaningful within the application's context.

13.2.3 Definitions, Applications, and Mitigation Strategies for Privacy and Security

AI systems typically need a very large amount of training data for algorithm development. Privacy concerns arise with the collection, handling, and usage of data. The best practice is to prioritize responsible data col-

lection and usage, in which users are fully aware of how data is collected, what it is used for, and how it may benefit individuals and society. The Cambridge Analytica incident, where users' data was harvested without consent for political advertising, was a major breach of privacy and exemplifies the risks to privacy in the age of AI (Graham-Harrison and Cadwalladr 2018). In order to safeguard privacy and security, AI applications should minimize the collection and use of data, anonymize data when appropriate, and restrict access to data. AI systems would adhere to robust cybersecurity and physical security protocols and data governance frameworks (Tilburg Law Review 2003; OCR 2021). Responsible data and systems governance should be centered around human rights and the well-being of society (Tomašev et al. 2020). Furthermore, implementing privacy-preserving ML techniques such as federated learning and differential privacy can ensure that AI systems learn from data without compromising individual privacy (Abadi et al. 2016).

As AI continues to shape the future of society and the environment, the ethical dimensions of bias and fairness, interpretability and explainability, and privacy and security demand our immediate attention. The potential risks are significant, but so are the benefits. Adopting comprehensive mitigation strategies and regulatory compliance can steer AI development toward enhancing societal welfare while upholding ethical standards.

13.3 AI Applications for Healthcare

A growing and aging global population presents new and increasing health challenges (Baker and Xiang 2023). AI presents an opportunity to advance toward one of the United Nations' SDGs: "ensuring healthy lives and promoting well-being for all at all ages" (SDGs 2016). Efforts to improve health and healthcare using AI include remote health monitoring and interventions, drug discovery and personalized care, medical imaging and diagnosis, and monitoring and predicting disease outbreaks and control. However, it is important to note that these technologies, if not developed and used ethically and responsibly, could potentially cause harm to individuals, local communities, and global society. This awareness is crucial to ensure the cautious and responsible use of AI in healthcare.

13.3.1 Remote Health Monitoring and Intervention

The internet of things (IoT) holds great promises to enhance healthcare standards within clinical settings and beyond. However, managing the vast amount of data these devices generate poses a significant challenge (Habibzadeh et al. 2020). This challenge has spurred the integration of AI into IoT systems, giving rise to the concept of artificial intelligence of things (AIoT) (Chang et al. 2021). AIoT represents a new era in healthcare, driving the transition from Healthcare 4.0 to Healthcare 5.0 (Mbunge et al. 2021), characterized by pervasive care that includes intelligent remote monitoring, personalized medicine, and emotive telemedicine. This paradigm shift aims to overcome the limitations of previous systems by leveraging advanced technologies within an AIoT architecture.

In health monitoring, AIoT is making strides, particularly in wearable technology. Wearables are widely used and are combined with novel methods for data extraction to generate substantial amounts of data (Sharifi et al. 2019; Baker et al. 2021a, 2021b; Nasarre et al. 2022). Sensor fusion approaches are being explored to synthesize data from multiple devices, showing promise in monitoring complex conditions such as physiological stress (Winslow et al. 2016) and pain (Winslow et al. 2022). Moreover, there is a growing interest in entirely noncontact monitoring systems, utilizing imagery and channel state information for assessing health parameters and activity levels (Khanam et al. 2021).

Advancements in diagnosis (Murugan et al. 2021; Rashed-Al-Mahfuz et al. 2021) and prognosis (Baker et al. 2020; Ghiasi et al. 2022) using AIoT have also been notable. Studies have achieved strong results by integrating data from wearable devices with demographic and clinical variables. However, challenges remain in achieving descriptive illness classification and reducing dependency on clinical parameters. Explainability techniques such as heat mapping and feature importance identification are prevalent but need validation to ensure genuine improvements in clinician trust (Saha et al. 2020).

Future research directions in AIoT include the development of advanced ML methods such as long short-term memory (LSTM)-based networks and transformers, the exploration of embedded and edge AI, data fusion for

diagnostics and prognostics, reducing dependency on clinical parameters, descriptive illness classification, and enhancing explainability tools. Standardized frameworks for validating and comparing explanation tools are recommended to ensure the responsible and effective implementation of AIoT systems in healthcare (Baker and Xiang 2023).

13.3.2 Drug Discovery and Personalized Medicine

The paradigm of drug discovery and personalized medicine has witnessed remarkable advancements in recent years, revolutionizing the approach to treating various diseases (Subbiah 2023). Conventional drug discovery methods have often proven to be time-intensive and financially burdensome, with many drugs failing to demonstrate adequate efficacy or causing undesirable side effects in specific patient populations (DiMasi et al. 2016). However, the emergence of innovative technologies powered by AI and a profound understanding of individual genetic variations has catalyzed a rapid advancement in drug discovery and personalized medicine (Schneider 2018; Deng et al. 2022).

Drug repurposing has emerged as a promising strategy, entailing the identification of existing drugs that can be employed to treat diseases beyond their original intent. Researchers can expedite the drug development process and mitigate associated costs by capitalizing on the established safety and pharmacokinetic data of already approved drugs. This approach has yielded successful outcomes, exemplified by the repurposing of minoxidil, initially developed for hypertension, to treat hair loss (Kirtonia et al. 2021; Xia et al. 2024).

Another groundbreaking advancement is the development of novel drugs by identifying and using the components of existing drugs with the desired effects. By isolating and investigating the specific fragments contributing to a drug's therapeutic effects, researchers can engineer new compounds that retain the desired properties while minimizing or eliminating adverse side effects. This targeted approach enhances efficacy and improves drug safety and tolerability (Sharma et al. 2016; Du et al. 2024).

Personalized medicine takes drug discovery to new heights by leveraging multimodal approaches to incorporate individual genetic variations, lifestyle, and other factors and their influence on drug response (Acosta et al. 2022). By analyzing a patient's genetic and overall profile, healthcare providers can customize drug treatments to optimize efficacy and minimize adverse reactions (Alowais et al. 2023). This approach has demonstrated remarkable success in oncology, where genetic testing can guide the selection of targeted therapies based on the unique molecular characteristics of a patient's tumor.

13.3.3 Epidemic Prediction and Management

The COVID-19 pandemic has posed significant challenges to global health systems, highlighting the need for innovative approaches to infectious disease detection and management. DL techniques have emerged as promising tools in various medical applications, including detecting thyroid issues, lung nodules, fetal localization, and diabetic retinopathy (Heidari et al. 2022). Given the diversity of clinical imaging sources such as MRI, X-ray, and CT scans, DL approaches appear well-suited to address the complexities of COVID-19. Numerous studies have explored DL's potential in combating the pandemic, employing techniques such as LSTM Networks (Demir 2021; Koç and Türkoğlu 2022), self-organizing maps (Melin et al. 2020; Galvan et al. 2021), convolutional neural networks (CNNs) (Abdel-Basset et al. 2021; Ismael and Şengür 2021), among others.

Several key strategies can be implemented to address the existing gaps in AI-based epidemiological research. First and foremost, enhancing data quality through improved collection and processing methods is crucial. This involves collaboration between AI experts and medical professionals to bridge knowledge gaps and foster interdisciplinary research (Al-Waisy et al. 2021). Robust DL techniques capable of handling noisy data and filtering out online rumors can significantly improve the effectiveness of AI models in pandemic analysis (ArunKumar et al. 2021).

Implementing appropriate regulatory frameworks is equally essential, balancing patient privacy with the deployment of AI in healthcare settings (Palaniappan et al. 2024). Investing in edge computing in-

frastructure and IoT devices can enhance data processing capabilities, facilitating real-time monitoring and analysis during pandemics (Ray et al. 2019).

Moreover, developing centralized data repositories for patient data could greatly aid predictive, diagnostic, and therapeutic approaches (Hruby et al. 2013). Benchmarking tools for assessing and comparing AI methodologies are vital for real-world applications, ensuring data transparency and interpretability. These measures collectively contributed to improving the reliability and effectiveness of AI solutions in combating COVID-19 and hold promise in addressing future pandemics (Heidari et al. 2022).

13.3.4 AI in Medical Imaging and Diagnostics

AI has revolutionized the field of medical imaging and diagnostics, offering unprecedented opportunities for early detection, accurate diagnosis, and personalized treatment of various diseases (Hosny et al. 2018; Alowais et al. 2023). One of the most promising developments in this area is using image foundation models for downstream tasks in medical imaging (Pai et al. 2024; Zhang and Metaxas 2024). These models, such as vision transformers (ViT) and self-supervised ViT, leverage large-scale pretraining on diverse image datasets to learn rich visual representations that can be fine-tuned for specific medical imaging tasks, such as segmentation and classification (Li and Rajpurkar 2024). Recent advancements in AI have also led to the development of more efficient models, such as Hyena and state space model. Hyena is a hierarchical vision transformer that combines the strengths of CNNs and transformers to achieve state-of-the-art performance on medical image segmentation tasks while being computationally efficient (Poli et al. 2023). Other notable models include Swin Transformer, which uses shifted window attention to capture local and global dependencies (Liu et al. 2021); UNet++, which incorporates deep supervision and skip connections (Zhou et al. 2018); nnFormer, a self-adapting framework for optimizing transformer-based models (Zhou et al. 2023); and 3D CNN, which directly processes volumetric medical imaging data (Wang 2023). These models efficiently handle large-scale, high-resolution medical imaging data while maintaining high performance in various tasks, such as segmentation, classification, and detec-

tion. The application of these cutting-edge AI models has shown remarkable success in the segmentation and diagnosis of various cancers and other diseases using medical imaging modalities such as CT, MRI, and PET scans. For instance, DL algorithms have demonstrated high accuracy in detecting and segmenting lung nodules, brain tumors, and breast lesions, facilitating early intervention and improving patient outcomes (Koh et al. 2022; Srivastava et al. 2023; Canayaz et al. 2024). Moreover, AI-based systems can assist radiologists in prioritizing critical cases, reducing diagnostic errors, and optimizing workflow efficiency (van Leeuwen et al. 2022). As AI advances, its integration into medical imaging and diagnostics is expected to grow exponentially, ultimately leading to more precise, personalized, and accessible healthcare solutions. However, addressing challenges such as data privacy, algorithmic bias, and regulatory frameworks is crucial to ensure AI's safe and ethical deployment in clinical settings.

13.3.5 Ethical Considerations in Healthcare AI

AI algorithms offer significant advancements in healthcare, including improvements in diagnosis and treatment by providing accurate and personalized treatment plans based on individual patient data, offering clinicians more informed decisions and recommendations, and detecting anomalies and patterns in the data to help both patients and clinicians with early disease detection (Elendu et al. 2023). AI-powered tools also improve patient care and outcomes through remote monitoring, robotic-assisted surgeries and devices, and advancements in the automation of administrative tasks, imaging analysis, and drug discovery (Elendu et al. 2023). With the growing applications of AI in the healthcare system, some ethical concerns regarding algorithm validation, interoperability, bias and discrimination, security, and patient privacy protections have arisen (Safdar et al. 2020). One primary concern is related to the misaligned and miscalibrated trust placed on automated systems (Azevedo-Sa et al. 2021). AI algorithms often perform more reliably with less bias and error toward the majority population, but they can fail when applied to underrepresented patient groups, leading to harmful consequences for these patients. Errors may be introduced by systematic collection of large amounts of data from inaccurate sensors or input devices (Hicks et al. 2019). Safeguarding against such consequences must be part of the AI al-

gorithm's design and deployment framework (Yale School of Medicine 2021). Due to the absence of clear guidelines regarding the legal and ethical implications of AI in healthcare, these concerns need further exploration (Immerse Education 2022; Elendu et al. 2023). It is essential for policymakers to actively address these ethical concerns to ensure that the benefits of AI outweigh the risks (Rigby 2019; Immerse Education 2022; Elendu et al. 2023). Ethical risks of AI can include data privacy and confidentiality, informed consent to use data, and security (Rigby 2019; Immerse Education 2022; Elendu et al. 2023). Data related to healthcare records and the history of patients are sensitive and always risk being shared for the wrong purposes without proper consent. Ensuring the privacy and security of this data is essential for patient trust and data integrity (Elendu et al. 2023). Healthcare organizations and other institutions that collect data from individuals need to implement responsible data collection and utilization procedures. This includes incorporating robust anonymization methods, encryption, and secure protocols for sharing data to protect patient confidentiality.

Algorithmic fairness and bias are other ethical concerns that need to be addressed. As mentioned earlier, AI algorithms can show bias toward certain populations, leading to inaccurate diagnosis and treatment. Transparency throughout the AI system lifecycle, from data collection, design, and development to deployment and explainability of system behavior, is required to establish adequate trust toward AI systems. For example, decision-making processes in healthcare AI must be transparent and explainable, enabling patients and healthcare providers to understand AI-driven recommendations better and make better decisions. Finally, it is essential to establish clear roles, responsibilities, and accountability for AI system users in healthcare. All ethical guidelines and frameworks should be actively updated and accessible to all stakeholders.

AI in healthcare has the potential to benefit both patients and healthcare providers if appropriately designed and utilized. By prioritizing ethical and responsible AI system design considerations and evaluating all societal implications of these technologies, we can ensure they enhance patient well-being and trust in the evolving healthcare landscape (Rigby 2019; Immerse Education 2022; Elendu et al. 2023).

13.4 AI for Environmental Sustainability

13.4.1 Introduction

Life on earth is increasingly in peril due to a changing climate, food insecurity, and a growing loss of biodiversity. These challenges are highlighted by the United Nations 17 Sustainable Development Goals (SDGs), 8 of which are focused on the water-energy-food nexus and environmental sustainability (SDGs 2016). Given the significant challenges facing humanity in this century (Warren et al. 2024), AI is being used in many applications to improve climate monitoring and modeling, promote sustainable agriculture, conserve biodiversity, and advance clean energy (Konya and Nematzadeh 2024).

13.4.2 AI for Climate Change Monitoring and Modeling

Individuals in regions highly vulnerable to climate change experience 15 times higher mortality rates from disasters compared to individuals in low-vulnerability regions (SDGs 2016). In an effort to better monitor and model climate change, AI models are being implemented to utilize vast amounts of historical and real-time data from satellite sensors, along with terrestrial, oceanic, and atmospheric weather stations, to monitor the earth from individual polar ice sheets (Braakmann-Folgmann et al. 2023) to worldwide changes in forest cover (Chen et al. 2023). AI-driven remote sensing (Ellis-Soto et al. 2023) is being used to track specific species of animals and plants critical to combating climate change and investigate impacts to biodiversity (Chowdhury et al. 2024). At a micro level, computer vision models combined with camera sensors are being used to analyze waste processing and recycling facilities to recycle more material (Greyparrot 2024), which is critical since waste is responsible for 16% of global greenhouse emissions (US EPA 2016). Leveraging a network of strategically placed cameras at rivers and on seafaring vessels, enhanced by satellite sensing, computer vision models are able to detect plastic waste in the world's waterways and enable large-scale cleanup efforts (Lebreton 2022).

13.4.3 AI for Sustainable Agriculture and Food Production

The United Nations has estimated that approximately one in three people worldwide struggle with moderate to severe food insecurity, and more than one in four people lack safe drinking water (SDGs 2016). Previous agricultural revolutions included the domestication of plants and animals to increase crop yield, the use of fertilizer, crop rotations, and mechanization to increase food production, and the genetic modification of crops to further increase food growth (Faris 2014). Several groups have argued that the application of AI technology is resulting in another agricultural revolution. In the face of an increasing world population, scarcity of resources, climate change, and a culture of food waste in richer countries, the fourth agricultural revolution is leveraging robotics, temperature and moisture sensors, aerial images, GPS technology, and AI to increase crop yield while also using a minimum of water and chemicals (De Clercq and Vats 2023). For instance, AI is being used to enable smart farming through planting; crop management, including weeding, thinning, pruning, plant protection measures; crop scouting through the use of sensors and geomapping; picking and harvesting; and postharvest technology, encompassing packaging, transport, and storage (Gupta and Gupta 2024). For instance, AI-enabled tractors are being deployed that integrate high-resolution imaging to distinguish weeds from crops with high-powered, high-precision lasers to eliminate weeds using thermal energy (Carbon Robotics 2024). Such an approach can function 24 hours a day autonomously while increasing crop yield and improving soil health without herbicides. Another emerging technology integrates high-resolution computer vision cameras with millimeter precision sprayers to identify crop status and apply herbicides, fertilizer, or pollen with a reduction of over 95% in the use of chemicals (Verdant Robotics 2024).

13.4.4 AI for Wildlife Conservation

The world is currently facing the largest extinction of species since the dinosaur age due to the combination of climate change, forest loss, land degradation, coastal eutrophication, ocean warming and acidification, plastic pollution, and overfishing (SDGs 2016). AI is increasingly being used in applications to analyze wildlife data at scale, predict changes to

species and their habitats, and inform policy to improve conservation efforts. For instance, hundreds of thousands of camera traps–digital cameras paired with infrared sensors that capture images when movement is detected – have been deployed worldwide, generating an enormous amount of data for researchers to manually catalog, label, and analyze (Kays et al. 2020). Leveraging cloud AI, one open-access tool uses AI for automated image labeling, including the presence of animals, determination of species or individual animal, geolocation, and time at scale, allowing for insights in animal migration, biodiversity pressure, and impacts on habitat (Ahumada et al. 2020). Other approaches combine mobile, high-resolution camera technology attached to drones with computer vision to detect and deter poaching, among other surveillance and security measures (Kamminga et al. 2018). Similarly, audio recordings in the wild are being used to identify species, specify interactions, and monitor migration (Kwok 2019).

13.4.5 AI for Renewable Energy Optimization

Over 675 million people do not have access to electricity, the majority of whom live in sub-Saharan Africa. While the transition away from fossil fuels continues, renewables, including wind, solar, geothermal, hydropower, and bioenergy, only represent a small amount of the energy being used worldwide for heating and transportation (SDGs 2016). AI is being used to better predict and forecast the availability of renewable energy sources, optimize electrical grids, and optimize the renewable supply chain (Jha et al. 2017). For instance, for power grids that leverage a combination of sources, AI has enabled smart control centers to optimize supply and demand resources (Omitaomu and Niu 2021). AI is also simplifying the transition to solar power at individual homes and businesses by automating solar planning, forecasting production, and estimating monetary savings (Bouquet et al. 2024). Ultimately, the integration of AI has greatly increased renewable energy efficiency and output, economic impact, and reduced its environmental impact (Ohalete et al. 2023).

13.4.6 Ethical Considerations in Environmental AI

While much progress has been made in applying AI to the environment, several ethical challenges have emerged. The increasing adoption of AI systems may lead to unprecedented economic and social implications, disrupting local and global job markets and economies (Vochozka et al. 2018). For instance, individuals working in the agricultural industry may face job loss due to the proliferation of AI-enabled robotics systems. The exponential increase in camera technology worldwide may enable increasingly sophisticated environmental modeling and wildlife conservation but may be misused by governments, leading to loss of privacy (Karpa et al. 2022). Additionally, the advent of AI could further exacerbate existing patterns of health inequities if the benefits of AI primarily support populations in high-income countries or privilege the wealthiest within countries (Murphy et al. 2021), which has the potential of reproducing and exacerbating unequal relationships of dependence and inequality amongst different populations around the world (Couldry and Mejias 2019).

13.5 AI for Education and Accessibility

Growing application of AI in education (AIED) has changed the landscape of education and led to improvement in student's learning progress (Kopp and Thomsen 2023). AI provides academic and nonacademic support to teachers and students with their personal and professional development goals. Academic support may include services such as personalized teaching and learning via tailored assessments, curriculum, or learning AI apps that can be personalized based on learners' engagement. AI has enabled new paradigms of innovations in education that were not possible within traditional classroom environments, for example, computer-supported group learning, interactive discussion forums, personalized learning experiences, and automated evaluation enhanced by facial recognition technology and predictive analysis. AI systems have also been utilized as educational policy-making advisors (Hwang et al. 2020). The integration of AI into education has profoundly transformed teaching and learning.

13.5.1 Personalized Learning with AI

The integration of AI into education aims to personalize learning experiences based on individual needs and abilities. It aligns with the goal of "ensuring inclusive and equitable quality education and promoting lifelong learning opportunities for all" (SDGs 2016). Traditional education systems often struggle to cater to the diverse learning requirements of students, leading to inefficiencies and disparities in learning outcomes. AI offers a solution by harnessing big data analysis to identify behavioral patterns and preferences, thereby enabling personalized recommendations and interventions (Al Hadwer et al. 2019).

Recent advancements in AI have paved the way for student-centered learning solutions, where the focus shifts from a one-size-fits-all approach to personalized learning experiences (Bhutoria 2022). AI systems can analyze learning abilities, habits, and preferences to recommend personalized learning paths and instructional strategies (Liang and Hainan 2019). This personalized approach not only improves learning outcomes but also enhances student engagement. Additionally, AI-powered diagnostic and predictive solutions enable proactive identification of learning difficulties and preemptive interventions, thereby reducing the risk of failure or dropout (Tong et al. 2019).

Furthermore, AI plays a significant role in education by creating adaptive learning materials, intelligent examination systems, and gamified learning environments (Mondal 2019; Li and Wang 2020). These innovations not only enhance skill development but also provide immersive learning experiences. Virtual and augmented reality simulations, for instance, offer a unique and engaging way for students to learn (Fan and Zhi 2020; Iwin Thanakumar Joseph et al. 2020).

AI holds immense potential to revolutionize education by personalizing learning experiences (Bhutoria 2022), enhancing learning outcomes, and fostering student engagement. However, the implementation of AI-enabled personalized learning poses challenges, such as the necessity for comprehensive data collection, infrastructure upgrades, and user training. Disparities in technological literacy and access to digital resources may amplify existing educational inequalities (Feng et al. 2019). Privacy

and security are critical aspects for building trust and mitigating potential risks associated with AI-powered systems (Tong et al. 2019). Overcoming these challenges is not a task for a single entity, but a collective responsibility. It requires collaborative efforts from policymakers, educators, and technologists, with each playing a crucial role, to ensure equitable access to AI-driven educational resources while safeguarding student privacy and security.

13.5.2 Ethical Aspects of AIED

There are ethical risks and concerns regarding the adaptation of AIED. Various international organizations, such as the International United Nations Educational, Scientific and Cultural Organization (UNESCO), the European Commission and Parliament, and the Organization for Economic Co-operation and Development (OECD), developed ethical guidelines to mitigate these risks and concerns. Six challenges to be addressed as outlined by UNESCO (2019) to ensure the practical and effective utilization of AIED are: (i) public policy, (ii) equity in AIED, (iii) equipping teachers for education powered by AI, (iv) training AI systems to comprehend educational contexts, (v) establishing robust data systems of high quality, and (vi) elevating the significance of research on ethical and transparent practices in data collection, utilization, and sharing (UNESCO 2019). These challenges, if left unaddressed have broad societal implications, such as expanding systemic bias and discrimination faced by underrepresented student populations and widening the gap of inequality among various vulnerable student groups and privacy concerns related to data collection and processing (Hwang et al. 2020; Holmes et al. 2022; Nguyen et al. 2023). Principles for responsible stewardship of trustworthy AI as outlined by OECD aim to achieve inclusiveness and human-centered approach to AI systems used in education. It also targets ethical handling and use of data and AI systems. These five principles are: (i) inclusive growth, sustainable development, and well-being; (ii) human-centered values and fairness; (iii) transparency and explainability; (iv) robustness, security, and safety; and (v) accountability (OECD 2024).

AI-powered education should be inclusive and equally accessible to all. Accessibility requires that the technology is affordable and user-friendly

for individuals from diverse demographics and cultures, and specifically those with disabilities. AI can break down barriers between learners and educational resources through technologies such as live captioning with multi-language and text-to-speech support. Such technologies can enable students with diverse learning needs and abilities to access educational materials (Van Laere and van Braak 2017; Remian 2019). Data collection efforts should include learners with disabilities as well as diverse learners. Accessibility also requires adequate internet and digital infrastructure as well as the personal digital devices. It's important to highlight that certain fundamental necessities, such as internet coverage and digital infrastructure, are crucial to achieve accessibility. Deployment of AIED should consider societal infrastructure and resources to incorporate a diverse range of individuals within the targeted region (Nguyen et al. 2023).

Ethical principles and concerns surrounding the use of AIED encompass governance to guarantee AI's appropriate application and ensure that the technology's design aligns with its intended objectives. This alignment aims to meet stakeholder needs while upholding accountability, equity, fairness, and individuals' privacy, security, and safety (Remian 2019; Borenstein and Howard 2021; Nguyen et al. 2023).

13.6 AI in Humanitarian Efforts and Disaster Response

Climate change has been linked to an increase in the frequency, intensity, and economic impact of natural disasters (Coronese et al. 2019), which are defined as phenomena that inflict harm through loss of life, damage to the environment, or economic loss and are beyond a community's ability to react (Abid et al. 2021). Natural disasters include severe weather, such as cold waves, drought, hail, heat waves, hurricanes, tornados, and tsunamis; floods, including coastal and riverine flooding; ground failure, such as avalanches, landslides, and earthquakes; and fire hazards, including volcanic activity and wildfires (FEMA 2023). Given the increasing impact of natural disasters, many AI applications have been developed to reinforce early warning systems, support disaster response and relief, and improve humanitarian crisis responses.

13.6.1 Early Warning Systems

The United Nations Office for Disaster Risk Reduction defines early warning systems (EWS) as an integrated system of hazard monitoring, forecasting and prediction, disaster risk assessment, communication, and preparedness activities (UNDRR 2007). ESW enables individuals, communities, governments, businesses, and others to take timely action to reduce disaster risks in advance of hazardous events. EWS consists of risk assessment algorithms combined with monitoring capabilities and warning communication tools (Esposito et al. 2022). EWS have been implemented worldwide to monitor and predict seasonal flooding using light detection and ranging sensors and river monitoring technologies (Wedajo 2017), to detect missile attacks using satellites (Podvig 2002), to provide early warning for earthquakes using seismic stations and wireless alert networks (Allen et al. 2009), and for preventing and mitigating disease outbreaks by assessing wastewater (Gahlot et al. 2023). However, regardless of growing international calls, relatively few countries have implemented national, multi-hazard EWS (UNDRR 2007). AI may improve the capabilities and accessibility of EWS by providing tools to sense, clean, process, and analyze data coming from disparate sources in the environment (Esposito et al. 2022). AI could also be used to revolutionize EWS by providing earlier warnings across a wider range of risks while also identifying and mitigating systemic risks in the systems (Wever et al. 2022).

13.6.2 Disaster Response and Relief

Disaster management consists of four phases: mitigation, focused on prevention; preparedness, focused on accumulating resources, implementing mutual aid agreements, and training; response, which focuses on allocating resources; and recovery, with the goal of minimizing adverse impacts (Sun et al. 2020). Mitigation and preparedness efforts are put in place prior to a disaster, response efforts are implemented during a disaster, and recovery efforts are initiated following the acute disaster phase. To date, the majority of AI applications in disaster management have focused on the disaster response phase (Sun et al. 2020). Examples of AI in disaster response include event mapping, damage assessment, resource allocation, inter-agency collaboration tools, and sentiment analysis (Sun et al.

2020). Such applications leverage a broad range of data, such as data from critical infrastructure, meteorological data, environmental data, imagery, remote sensing, telecommunications, and social media, among others (Cao 2023). AI techniques leverage such disparate data sources and manipulate them using ML and DL techniques to determine patterns (Linardos et al. 2022). For example, fuzzy neural networks have been used to better predict precipitation during severe storms in China (Huang et al. 2018), Naïve Bayes trees have been used to predict landslide susceptibility in Rwanda (Nsengiyumva and Valentino 2020), natural language processing has improved disaster monitoring through social media monitoring (Domala et al. 2020), and various DL approaches have been implemented with UAVs to support search and rescue (Bejiga et al. 2017).

13.6.3 Humanitarian Crisis Response

Humanitarian crises are an event or series of events that represent a critical threat to the health, safety, security, or well-being of a community or large group of people, usually over a wide area (Humanitarian Coalition 2024). In addition to natural disasters already described, humanitarian crises can be man-made, resulting from armed conflicts and associated violence, loss of life, and displacement of populations; plane or train crashes; and epidemics, among others.

An increasingly important source of data during humanitarian crises comes from user-generated content available on social media, including text, images, audio, and video (Haworth and Bruce 2015). While social media helps confront threats and keeps people connected, large-scale analysis of such data using DL, natural language processing, and computer vision also allows for rapid and precise classification of social media messages during crises (Padhee et al. 2020). Other data sources include meteorological data and market analytics, such as the forecast-based financing system of the International Federation of Red Cross and Red Crescent Societies, which enables swift allocation of resources for early action in humanitarian emergencies (Couldry and Mejias 2019; Lohrey 2020). AI has the potential to provide a paradigm shift in humanitarian crises by moving from reactive to anticipatory approaches (Beduschi 2022). For example, Project Jetson is an initiative from the UN High

Commissioner for Refugees, which combines AI, design techniques, and qualitative research methods to forecast the forced displacement of people (UNHCR 2019). The World Food Programme is using predictive analytics to forecast food insecurity in conflict zones using data on current hazards, conflicts, food consumption, vegetation, and rainfall (WFP 2024). The UN Rapid Mapping Service applies AI to satellite imagery to assess damage caused by conflict or natural disasters to inform humanitarian responses on the ground (Vacarelu and Aylett-Bullock 2021).

13.6.4 Ensuring Fairness and Transparency in AI for Humanitarian Efforts

Many existing and emerging AI applications in humanitarian efforts have not yet been fully tested, and a fair question is whether or not such models will lead to better preparedness and responses. Many models are trained on existing data and may not be the best solutions for novel crises. For instance, many models leveraged in the COVID-19 crisis failed to provide sufficient support to medical decision-making due to low-quality or inexistent training data and a high risk of bias (Wynants et al. 2020). Big data approaches for anticipating Ebola outbreaks in West Africa have not been as effective as investing in public health and social infrastructure (Wamsley and Chin-Yee 2021). Similarly, social media data may be scarce, evasive, or worse, deceptive during armed conflicts due to active disinformation campaigns (Udupa et al. 2020). Due to these concerns and others, humanitarian actors should disclose information about what data is being used, for how long, how it will be stored, and who will have access to it to ensure transparency (Beduschi 2022). Additionally, a governance framework that draws from human rights and incorporates ethics may ensure that AI is used for humanitarian efforts in ways that preserve fairness and transparency (Pizzi et al. 2020).

13.7 Conclusion

Efforts to achieve the SDGs are at crossroads, and AI presents new opportunities for the global community to act (United States Department of State 2023). Working across industries, universities, governments, and civil society organizations, AI should be developed and deployed respon-

sibly to drive economic growth, reduce poverty and inequality, advance environmental stability, improve people's lives, and empower individuals in all societies across all stages of development (Arya and Binu 2024). There are myriad emerging applications that can support humans and the environment (Table 13.1).

TABLE 13.1

How AI is supporting the SDGs.

Sustainable development goal	How AI is being or can be applied
1. End poverty in all its forms everywhere	Forecast food insecurity using data on consumption, vegetation, and rainfall (WFP 2024)
2. End hunger, achieve food security and improved nutrition, and promote sustainable agriculture	Smart farming for increased crop yield and decreased resources used (Gupta and Gupta 2024)
3. Ensure healthy lives and promote well-being for all at all ages	
4. Ensure inclusive and equitable quality education and promote lifelong learning opportunities for all	AI is revolutionizing education by offering personalized, student-centered systems and predictive diagnostic tools that cater to all learning styles (Bhutoria 2022)
5. Achieve gender equality and empower all women and girls	Training AI models to avoid gender, culture, religion, and other forms of discrimination (Patón-Romero et al. 2022)

Sustainable development goal	How AI is being or can be applied
6. Ensure availability and sustainable management of water and sanitation for all	Analyze sensor data to monitor water quality, usage patterns, and optimization (Rana et al. 2023)
7. Ensure access to affordable, reliable, sustainable, and modern energy for all	Better modeling and use of existing resources (Jha et al. 2017)
8. Promote sustained, inclusive, and sustainable economic growth, full and productive employment, and decent work for all	Match job seekers with suitable opportunities, tailor learning opportunities, streamline processes, and improve productivity (Arya and Binu 2024)
9. Build resilient infrastructure, promote inclusive and sustainable industrialization, and foster innovation	AI algorithms for threat detection, risk assessment, and automated responses to protect infrastructure (Sarker 2024)
10. Reduce inequality within and among countries	Enhance efficiency, accessibility, and affordability across sectors to reduce disparities in access to resources (Farahani and Ghasemi 2024)

Sustainable development goal	How AI is being or can be applied
11. Make cities and human settlements inclusive, safe, resilient, and sustainable	Improving waste management, air quality monitoring, disaster response, and transportation in cities (Gupta and Degbelo 2023)
12. Ensure sustainable consumption and production patterns	Improve coordination between production and consumption while reducing carbon footprint (Chandra and Verma 2023)
13. Take urgent action to combat climate change and its impacts	Better climate monitoring and combating climate change (Cowls et al. 2023)
14. Conserve and sustainably use the oceans, seas, and marine resources for sustainable development	Ocean health monitoring (Mandal and Ghosh 2024) and cleanup (Lebreton 2022)
15. Protect, restore, and promote sustainable use of terrestrial ecosystems, sustainably manage forests, combat desertification, and halt and reverse land degradation and halt biodiversity loss	Global monitoring of species for preservation (Ahumada et al. 2020); autonomous vehicles for forest management (Ahumada et al. 2020; Buchelt et al. 2024); modeling of shifts to terrestrial ecosystems (Sun et al. 2023)

Sustainable development goal	How AI is being or can be applied
16. Promote peaceful and inclusive societies for sustainable development, provide access to justice for all, and build effective, accountable, and inclusive institutions at all levels	Supporting conflict resolution, economic stability, human rights protection, and justice (Dasandi and Jankin 2019)
17. Strengthen the means of implementation and revitalize the global partnership for sustainable development	Improved data sharing and collaboration among organizations, countries, and regions to address global challenges (Arya and Binu 2024)

Supporting the vision of AI for good requires AI that is fair and transparent to users, ethical, and respectful of the privacy of both individuals, which contribute to the underlying data and end users of the AI. Ensuring that AI is implemented, which respects human values and promotes human well-being also requires careful algorithm and UI design and oversight from individual developers and industries to sectors and global organizations. We echo calls for the products of AI development to be compatible with human values (Ozmen Garibay et al. 2023). AI that is centered in human values will support the major challenges facing humanity and provide a future of prosperity, fairness, and well-being for all.

References

Abadi, M., Chu, A., Goodfellow, I., et al. (2016). Deep learning with differential privacy. In *Proceedings of the 2016 ACM SIGSAC Conference on*

Computer and Communications Security (CCS '16), pp. 308–318. NY, USA: Association for Computing Machinery.

Abdel-Basset, M., Chang, V., Hawash, H., et al. (2021). FSS-2019-nCov: a deep learning architecture for semi-supervised few-shot segmentation of COVID-19 infection. *Knowledge-Based Systems* 212: 106647.

Abid, S.K., Noralifishah, S., Shiau, W.C., et al. (2021). Toward an integrated disaster management approach: how artificial intelligence can boost disaster management. *Sustainability: Science Practice and Policy* 13 (22): 12560.

Acosta, J.N., Falcone, J.F., Rajpurkar, P., et al. (2022). Multimodal biomedical AI. *Nature Medicine* 28 (9): 1773–1784.

Ahumada, J.A., Fegraus, E., Birch, T., et al. (2020). Wildlife insights: a platform to maximize the potential of camera trap and other passive sensor wildlife data for the planet. *Environmental Conservation* 47 (1): 1–6.

AI Explainability 360 (2023). *AI explainability 360*. https://aix360.res.ibm.com/ (accessed 8 June 2024).

Al Hadwer, A., Gillis, D., and Rezania, D. (2019). Big data analytics for higher education in the cloud era. In *2019 IEEE 4th International Conference on Big Data Analytics (ICBDA)* (pp. 203–207). IEEE.

Al-Waisy, A.S., Mohammed, M.A., Al-Fahdawi, S., et al. (2021). COVID-DeepNet: hybrid multimodal deep learning system for improving COVID-19 pneumonia detection in chest X-ray images. *Computers, Materials & Continua* 67 (2): 2409–2429.

Allen, R.M., Gasparini, P., Kamigaichi, O., et al. (2009). The status of earthquake early warning around the world: an introductory overview. *Seismological Research Letters* 80 (5): 682–693.

Alowais, S.A., Alghamdi, S.S., Alsuhebany, N., et al. (2023). Revolutionizing healthcare: the role of artificial intelligence in clinical practice. *BMC Medical Education* 23 (1): 689.

Andrews, E.L. (2021). *How flawed data aggravates inequality in credit*. Stanford HAI. https://hai.stanford.edu/news/how-flawed-data-aggravates-inequality-credit (accessed 24 June 2024).

Apley, D.W. and Zhu, J. (2020). Visualizing the effects of predictor variables in black box supervised learning models. *Journal of the Royal Statistical Society. Series B, Statistical methodology* 82 (4): 1059–1086.

Artificial Intelligence Index (2024). *AI index report 2024*. https://aiindex.stanford.edu/report/ (accessed 20 June 2024).

ArunKumar, K.E., Kalaga, D.V., Kumar, C.M.S., et al. (2021). Forecasting of COVID-19 using deep layer recurrent neural networks (RNNs) with gated recurrent units (GRUs) and long short-term memory (LSTM) cells. *Chaos, Solitons, and Fractals* 146: 110861.

Arya, M.K.J. and Binu, V.P. (2024). Helping hand of AI for achieving SDG. *Journal of Applied Science, Engineering, Technology and Management* 2 (01): 29–34.

Azevedo-Sa, H., Hebert, Y., Xi, J., et al. (2021). Using trust in automation to enhance driver-(semi)autonomous vehicle interaction and improve team performance. *arXiv [cs.RO]*. http://arxiv.org/abs/2106.02136.

Baker, S. and Xiang, W. (2023). Artificial intelligence of things for smarter healthcare: a survey of advancements, challenges, and opportunities. *IEEE Communications Surveys* & *Tutorials* 25 (2): 1261–1293.

Baker, S., Xiang, W., and Atkinson, I. (2020). Continuous and automatic mortality risk prediction using vital signs in the intensive care unit: a hybrid neural network approach. *Scientific Reports* 10 (1): 21282.

Baker, S., Xiang, W., and Atkinson, I. (2021a). A hybrid neural network for continuous and non-invasive estimation of blood pressure from raw electrocardiogram and photoplethysmogram waveforms. *Computer Methods and Programs in Biomedicine* 207: 106191.

Baker, S., Xiang, W., and Atkinson, I. (2021b). Determining respiratory rate from photoplethysmogram and electrocardiogram signals using respiratory quality indices and neural networks. *PLoS One* 16 (4): e0249843.

Beduschi, A. (2022). Harnessing the potential of artificial intelligence for humanitarian action: opportunities and risks. *International Review of the Red Cross* 104 (919): 1149–1169.

Bejiga, M.B., Zeggada, A., Abdelhamid, N., et al. (2017). A convolutional neural network approach for assisting avalanche search and rescue operations with UAV imagery. *Remote Sensing* 9 (2): 100.

Bhatt, U., Xiang, A., Sharma, S., et al. (2020). Explainable machine learning in deployment. In *Proceedings of the 2020 Conference on Fairness, Accountability, and Transparency (FAT* '20)*, pp. 648–657. NY, USA: Association for Computing Machinery.

Bhutoria, A. (2022). Personalized education and artificial intelligence in the United States, China, and India: a systematic review using a hu-

man-in-the-loop model. *Computers and Education: Artificial Intelligence* 3: 100068.

Biecek, P. (2018). DALEX: explainers for complex predictive models in R. *Journal of Machine Learning Research* 19 (84): 1–5.

Blatt, J. (2024). *Advances in facial recognition technology have outpaced laws, regulations; new report recommends federal government take action on privacy, equity, and civil liberties concerns*. National Academies. https://www.nationalacademies.org/news/2024/01/advances-in-facial-recognition-technology-have-outpaced-laws-regulations-new-report-recommends-federal-government-take-action-on-privacy-equity-and-civil-liberties-concerns (accessed 24 June 2024).

Bogen, M. (2019). All the ways hiring algorithms can introduce bias. *Harvard Business Review*. https://hbr.org/2019/05/all-the-ways-hiring-algorithms-can-introduce-bias (accessed 24 June 2024).

Borenstein, J. and Howard, A. (2021). Emerging challenges in AI and the need for AI ethics education. *AI and Ethics* 1 (1): 61–65.

Bouquet, P., Jackson, I., Mostafa, N., et al. (2024). AI-based forecasting for optimised solar energy management and smart grid efficiency. *International Journal of Production Research* 62 (13): 4623–4644.

Braakmann-Folgmann, A., Shepherd, A., Hogg, D., et al. (2023). Mapping the extent of giant Antarctic icebergs with deep learning. *The Cryosphere* 17 (11): 4675–4690.

Buchelt, A., Adrowitzer, A., Kieseberg, P., et al. (2024). Exploring artificial intelligence for applications of drones in forest ecology and management. *Forest Ecology and Management* 551: 121530.

Canayaz, M., Sanem Şehribanoğlu, S., Özgökçe, M., et al. (2024). A comprehensive exploration of deep learning approaches for pulmonary nodule classification and segmentation in chest CT images. *Neural Computing & Applications* 36 (13): 7245–7264.

Cao, L. (2023). AI and data science for smart emergency, crisis and disaster resilience. *International Journal of Data Science and Analytics* 15 (3): 231–246.

Carbon Robotics (2024). *LaserWeeder implement*. https://carbonrobotics.com/laserweeder (accessed 8 June 2024).

Castelnovo, A., Crupi, R., Greco, G., et al. (2022). A clarification of the nuances in the fairness metrics landscape. *Scientific Reports* 12 (1): 4209.

Caton, S. and Haas, C. (2024). Fairness in machine learning: a survey. *ACM Computing Surveys* 56 (7): 1–38.

Center for Security and Emerging Technology (CSET) (2021). *Ethical norms for new generation artificial intelligence released*. https://cset.georgetown.edu/publication/ethical-norms-for-new-generation-artificial-intelligence-released/ (accessed 8 June 2024).

Chandra, S. and Verma, S. (2023). Big data and sustainable consumption: a review and research agenda. *Vision Research* 27 (1): 11–23.

Chang, Z., Liu, S., Xingxing, X., et al. (2021). A survey of recent advances in edge-computing-powered artificial intelligence of things. *IEEE Internet of Things Journal* 8 (18): 13849–13875.

Chen, M., Dong, W., Yu, H., et al. (2023). Multimodal deep learning for mapping forest dominant height by fusing GEDI with earth observation data. *arXiv [cs.CV]*. http://arxiv.org/abs/2311.11777.

Chen, Z. (2023). Ethics and discrimination in artificial intelligence-enabled recruitment practices. *Humanities and Social Sciences Communications* 10 (1): 1–12.

Chin-Rothmann, C. and Lee, N. T. (2022). *Police surveillance and facial recognition: why data privacy is imperative for communities of color*. https://www.brookings.edu/articles/police-surveillance-and-facial-recognition-why-data-privacy-is-an-imperative-for-communities-of-color/ (accessed 24 June 2024).

Chowdhury, M., Martinez-Sansigre, A., Mole, M., et al. (2024). AI-driven remote sensing enhances Mediterranean seagrass monitoring and conservation to combat climate change and anthropogenic impacts. *Scientific Reports* 14 (1): 8360.

Coronese, M., Lamperti, F., Keller, K., et al. (2019). Evidence for sharp increase in the economic damages of extreme natural disasters. *Proceedings of the National Academy of Sciences of the United States of America* 116 (43): 21450–21455.

Couldry, N. and Mejias, U.A. (2019). Data colonialism: rethinking big data's relation to the contemporary subject. *Television & New Media* 20 (4): 336–349.

Cowls, J., Tsamados, A., Taddeo, M., et al. (2023). The AI gambit: leveraging artificial intelligence to combat climate change-opportunities, challenges, and recommendations. *AI & Society* 38 (1): 283–307.

Dasandi, N. and Jankin, S. (2019). *AI for SDG-16 on peace, justice, and strong institutions: Tracking progress and assessing impact*. https://sjankin.com/assets/img/research/ijcai19-sdg16.pdf.

De Clercq, M. and Vats, A. (2023). *Agriculture 4.0 – farming technology to meet food and climate goals*. https://www.marshmclennan.com/insights/publications/2018/mar/agriculture-4-0.html (accessed 8 June 2024).

Demir, F. (2021). DeepCoroNet: a deep LSTM approach for automated detection of COVID-19 cases from chest X-ray images. *Applied Soft Computing* 103: 107160.

Deng, J., Yang, Z., Ojima, I., et al. (2022). Artificial intelligence in drug discovery: applications and techniques. *Briefings in Bioinformatics* 23 (1): bbab430. https://doi.org/10.1093/bib/bbab430.

Dhurandhar, A., Chen, P., Luss, R., et al. (2018). Explanations based on the missing: towards contrastive explanations with pertinent negatives. In *Proceedings of the 32nd International Conference on Neural Information Processing Systems (NIPS'18), Red Hook*, pp. 590–601. NY, USA: Curran Associates Inc.

DiMasi, J.A., Grabowski, H.G., and Hansen, R.W. (2016). Innovation in the pharmaceutical industry: new estimates of R&D costs. *Journal of Health Economics* 47: 20–33.

Dogheim, G.M. and Hussain, A. (2023). Patient care through AI-driven remote monitoring: analyzing the role of predictive models and intelligent alerts in preventive medicine. *Journal of Contemporary Healthcare Analytics* 7 (1): 94–110.

Domala, J., Dogra, M., Vinit Masrani, V., et al. (2020). Automated identification of disaster news for crisis management using machine learning and natural language processing. In *2020 International Conference on Electronics and Sustainable Communication Systems (ICESC)* (pp. 503–508). IEEE.

Du, Y., Jamasb, A.R., Guo, J., et al. (2024). Machine learning-aided generative molecular design. *Nature Machine Intelligence* 6: 1–16.

Elendu, C., Amaechi, D.C., Elendu, T.C., et al. (2023). Ethical implications of AI and robotics in healthcare: a review. *Medicine* 102 (50): e36671.

Ellis-Soto, D., Wikelski, M., and Jetz, W. (2023). Animal-borne sensors as a biologically informed lens on a changing climate. *Nature Climate Change* 13 (10): 1042–1054.

Esposito, M., Palma, L., Belli, A., et al. (2022). Recent advances in internet of things solutions for early warning systems: a review. *Sensors* 22 (6): 2124. https://doi.org/10.3390/s22062124.

EU Artificial Intelligence Act (2024). *EU Artificial Intelligence Act*. https://artificialintelligenceact.eu/ (accessed 8 June 2024).

Fan, J. and Zhi, L. (2020). Design and implementation of virtual immersive classroom in big data environment. In *2020 15th International Conference on Computer Science & Education (ICCSE)* (pp. 430–432). IEEE.

Farahani, M.S. and Ghasemi, G. (2024). *Artificial intelligence and inequality: challenges and opportunities*. *Qeios*. https://doi.org/10.32388/7hwuz2.

Faris, J.D. (2014). Wheat domestication: key to agricultural revolutions past and future. In *Genomics of plant genetic resources*: Volume 1. *Managing, Sequencing and Mining Genetic Resources* (ed. R. Tuberosa, A. Graner, and E. Frison), 439–464. Dordrecht: Springer Netherlands.

Federal Emergency Management Agency (FEMA) (2023). *Natural hazards*. https://hazards.fema.gov/nri/natural-hazards (accessed 10 June 2024).

Feng, Y., Zhan, G., and Li, Z. (2019). Research on the co-construction and sharing of teaching resources under the "Internet Plus" environment. In *2019 10th International Conference on Information Technology in Medicine and Education (ITME)* (pp. 498–502). IEEE.

Friedman, J.H. (2001). Greedy function approximation: a gradient boosting machine. *The Annals of Statistics* 29 (5): 1189–1232.

Gahlot, P., Alley, K.D., Arora, S., et al. (2023). Wastewater surveillance could serve as a pandemic early warning system for COVID–19 and beyond. *WIREs Water* 10 (4): e1650. https://doi.org/10.1002/wat2.1650.

Galvan, D., Effting, l., Cremasco, H., et al. (2021). The spread of the COVID-19 outbreak in Brazil: an overview by Kohonen self-organizing map networks. *Medicina* 57 (3): 235. https://doi.org/10.3390/medicina57030235.

Garcia, A.C.B., Garcia, M.G.P., and Rigobon, R. (2023). Algorithmic discrimination in the credit domain: what do we know about it? *AI & Society* 39: 2059–2098. https://doi.org/10.1007/s00146-023-01676-3.

Ghiasi, S., Zhu, T., Lu, P., et al. (2022). Sepsis mortality prediction using wearable monitoring in low-middle income countries. *Sensors* 22 (10): 3866. https://doi.org/10.3390/s22103866.

Goldstein, A., Kapelner, A., Bleich, J., et al. (2015). Peeking inside the black box: visualizing statistical learning with plots of individual conditional expectation. *Journal of Computational and Graphical Statistics* 24 (1): 44–65.

GOV.UK (2021). *The roadmap to an effective AI assurance ecosystem*. https://www.gov.uk/government/publications/the-roadmap-to-an-effective-ai-assurance-ecosystem (accessed 8 June 2024).

Graham-Harrison, E. and Cadwalladr, C. (2018). Revealed: 50 million Facebook profiles harvested for Cambridge Analytica in major data breach. *The Guardian* https://www.theguardian.com/news/2018/mar/17/cambridge-analytica-facebook-influence-us-election (accessed 8 June 2024).

Greenwell, B.M., Boehmke, B.C., and McCarthy, A.J. (2018). A simple and effective model-based variable importance measure. *arXiv [stat.ML]*. http://arxiv.org/abs/1805.04755.

Greyparrot AI (2024). *Unlock the power of AI waste analytics*. https://www.greyparrot.ai/ (accessed 8 June 2024).

Guidotti, R., Monreale, A., Ruggieri, S., et al. (2018). A survey of methods for explaining black box models. *ACM Computing Surveys* 51 (5): 1–42.

Gupta, N. and Gupta, P.K. (2024). Robotics and artificial intelligence (AI) in agriculture with major emphasis on food crops. In *Digital Agriculture: A Solution for Sustainable Food and Nutritional Security* (ed. P.M. Priyadarshan, et al.), 577–605. Cham: Springer International Publishing.

Gupta, S. and Degbelo, A. (2023). An empirical analysis of AI contributions to sustainable cities (SDG 11). In: *The Ethics of Artificial Intelligence for the Sustainable Development Goals* (ed. F. Mazzi and L. Floridi), 461–484. Cham: Springer International Publishing.

Habibzadeh, H., Dinesh, K., Shishvan, O.R., et al. (2020). A survey of healthcare internet-of-things (HIoT): a clinical perspective. *IEEE Internet of Things Journal* 7 (1): 53–71.

Hale, K. (2021). A.I. bias caused 80% of black mortgage applicants to be denied. *Forbes Magazine*. https://www.forbes.com/sites/korihale/2021/09/02/ai-bias-caused-80-of-black-mortgage-applicants-to-be-denied/ (accessed 24 June 2024).

Haworth, B. and Bruce, E. (2015). A review of volunteered geographic information for disaster management. *Geography Compass* 9 (5): 237–

250.

Heidari, A., Navimipour, N.J., Unal, M., et al. (2022). The COVID-19 epidemic analysis and diagnosis using deep learning: a systematic literature review and future directions. *Computers in Biology and Medicine* 141: 105141.

Hicks, J.L., Althoff, T., Sosic, R., et al. (2019). Best practices for analyzing large-scale health data from wearables and smartphone apps. *NPJ Digital Medicine* 2: 45.

Holmes, W., Porayska-Pomsta, K., Holstein, K., et al. (2022). Ethics of AI in education: towards a community-wide framework. *International Journal of Artificial Intelligence in Education* 32 (3): 504–526.

Holzinger, A., Biemann, C., Pattichis, C.S., et al. (2017). What do we need to build explainable AI systems for the medical domain? *arXiv [cs.AI]*. http://arxiv.org/abs/1712.09923.

Hosny, A., Parmar, C., Quackenbush, J., et al. (2018). Artificial intelligence in radiology. *Nature Reviews Cancer* 18 (8): 500–510.

Hruby, G.W., McKiernan, J., Suzanne Bakken, S., et al. (2013). A centralized research data repository enhances retrospective outcomes research capacity: a case report. *Journal of the American Medical Informatics Association* 20 (3): 563–567.

Huang, Y., Jin, L., Zhao, H., et al. (2018). Fuzzy neural network and LLE Algorithm for forecasting precipitation in tropical cyclones: comparisons with interpolation method by ECMWF and stepwise regression method. *Natural Hazards* 91 (1): 201–220.

Humanitarian Coalition (2024). *What is a humanitarian emergency*? https://www.humanitariancoalition.ca/what-is-a-humanitarian-emergency (accessed 14 June 2024).

Hwang, G.-J., Xie, H., Wah, B.W., et al. (2020). Vision, challenges, roles and research issues of artificial intelligence in education. *Computers and Education: Artificial Intelligence* 1: 100001.

Immerse Education (2022). *Ethical issues with artificial intelligence and healthcare*. Immerse Education. https://www.immerse.education/study-tips/ethical-issues-with-artificial-intelligence-and-healthcare/ (accessed 20 June 2024).

Infocomm Media Development Authority (2024). *Artificial intelligence in Singapore*. https://www.imda.gov.sg/about-imda/emerging-technologies-and-research/artificial-intelligence (accessed 8 June 2024).

Innovation, Science and Economic Development Canada (2023). *Artificial Intelligence and Data Act*. https://ised-isde.canada.ca/site/innovation-better-canada/en/artificial-intelligence-and-data-act (accessed 8 June 2024).

InterpretML (2023). *InterpretML*. http://interpret.ml/ (accessed 8 June 2024).

Ismael, A.M. and Şengür, A. (2021). Deep learning approaches for COVID-19 detection based on chest X-ray images. *Expert Systems with Applications* 164: 114054.

Iwin Thanakumar Joseph, S., Benson Edwin Raj, S., and Kiyasudeen, J.M. (2020). Virtual reality—a paradigm shift in education pedagogy. In *2020 7th International Conference on Information Technology Trends (ITT)* (pp. 72–79). IEEE.

Jha, S. K., Xie, H., Wah, B.W., et al. (2017). Renewable energy: present research and future scope of Artificial Intelligence. *Renewable and Sustainable Energy Reviews* 77: 297–317.

Kamminga, J., Ayele, E., Meratnia, N., et al. (2018). Poaching detection technologies—a survey. *Sensors* 18 (5): 1474. https://doi.org/10.3390/s18051474.

Karpa, D., Klarl, T., and Rochlitz, M. (2022). Artificial intelligence, surveillance, and big data. In *Diginomics Research Perspectives: The Role of Digitalization in Business and Society* (ed. L. Hornuf), 145–172. Springer International Publishing.

Kays, R., Arbogast, B., Baker-Whatton, M., et al. (2020). An empirical evaluation of camera trap study design: how many, how long and when? *Methods in Ecology and Evolution* 11 (6): 700–713.

Khanam, F.-T.-Z., Perera, A.G., Al-Naji, A., et al. (2021). Non-contact automatic vital signs monitoring of infants in a neonatal intensive care unit based on neural networks. *Journal of Imaging* 7 (8): 122. https://doi.org/10.3390/jimaging7080122.

Kirtonia, A., Gala, K., Fernandes, S.G., et al. (2021). Repurposing of drugs: an attractive pharmacological strategy for cancer therapeutics. *Seminars in Cancer Biology* 68: 258–278.

Koç, E. and Türkoğlu, M. (2022). Forecasting of medical equipment demand and outbreak spreading based on deep long short-term memory network: the COVID-19 pandemic in Turkey. *Signal, Image, and Video Processing* 16 (3): 613–621.

Koh, D.-M., Papanikolaou, N., Bick, U., et al. (2022). Artificial intelligence and machine learning in cancer imaging. *Communication & Medicine* 2: 133.

Konya, A. and Nematzadeh, P. (2024). Recent applications of AI to environmental disciplines: a review. *The Science of the Total Environment* 906: 167705.

Kopp, W. and Thomsen, B.S. (2023). *How AI can accelerate students' holistic development and make teaching more fulfilling*. World Economic Forum. https://www.weforum.org/agenda/2023/05/ai-accelerate-students-holistic-development-teaching-fulfilling/ (accessed 20 June 2024).

Kwok, R. (2019). AI empowers conservation biology. *Nature* 567 (7746): 133–134.

Lebreton, L. (2022). The status and fate of oceanic garbage patches. *Nature Reviews Earth & Environment* 3 (11): 730–732.

Liang, Q. and Hainan, N.C. (2019). Adaptive learning model and implementation based on big data. In *2019 2nd International Conference on Artificial Intelligence and Big Data (ICAIBD)* (pp. 183–186). IEEE.

Li, K. and Rajpurkar, P. (2024). *Adapting segment anything models to medical imaging via fine-tuning without domain pretraining*. https://openreview.net/pdf?id=Fxi7pRmnYJ (accessed 24 June 2024).

Li, H. and Wang, H. (2020). Research on the application of artificial intelligence in education. In *2020 15th International Conference on Computer Science & Education (ICCSE)* (pp. 589–591). IEEE.

Linardos, V., Drakaki, M., Tzionas, P., et al. (2022). Machine learning in disaster management: recent developments in methods and applications. *Machine Learning and Knowledge Extraction* 4 (2): 446–473.

Liu, Z., Lin, Y., Cao, Y., et al. (2021). Swin transformer: hierarchical vision transformer using shifted windows. In *2021 IEEE/CVF International Conference on Computer Vision (ICCV)* (pp. 9992–10002). IEEE.

Lohrey, S. (2020). *Forecast-based financing*. https://www.forecast-based-financing.org/ (accessed 14 June 2024).

Lundberg, S.M. and Lee, S.-I. (2017). A unified approach to interpreting model predictions. *Advances in Neural Information Processing Systems* (pp. 4765–4774).

Makhlouf, K., Zhioua, S., and Palamidessi, C. (2021). On the applicability of machine learning fairness notions. *SIGKDD Explorations Newsletter* 23 (1): 14–23.

Mandal, A. and Ghosh, A.R. (2024). AI-driven surveillance of the health and disease status of ocean organisms: a review. *Aquaculture International: Journal of the European Aquaculture Society* 32 (1): 887–898.

Mbunge, E., Muchemwa, B., Jiyane, S., et al. (2021). Sensors and healthcare 5.0: transformative shift in virtual care through emerging digital health technologies. *Global Health Journal* 5 (4): 169–177.

Mehrabi, N., Morstatter, F., Saxena, N., et al. (2021). A survey on bias and fairness in machine learning. *ACM Computing Surveys* 54 (6): 1–35.

Melin, P., Monica, J.C., Sanchez, D., et al. (2020). Analysis of spatial spread relationships of coronavirus (COVID-19) pandemic in the world using self organizing maps. *Chaos, Solitons & Fractals* 138: 109917.

Milberg, T. (2024). *The future of learning: AI is revolutionizing education 4.0*. World Economic Forum. https://www.weforum.org/agenda/2024/04/future-learning-ai-revolutionizing-education-4-0/ (accessed 20 June 2024).

Ministry of Economy, Trade and Industry (METI) (2022). *"Governance guidelines for implementation of AI principles Ver. 1.1" Compiled*. https://www.meti.go.jp/english/press/2022/0128_003.html (accessed 8 June 2024).

Mitchell, S., Potash, E., Barocas, S., et al. (2021). Algorithmic fairness: choices, assumptions, and definitions. *Annual Review of Statistics and its Application* 8: 141–163.

Mondal, K. (2019). A synergy of artificial intelligence and education in the 21st century classrooms. In *2019 International Conference on Digitization (ICD)* (pp. 68–70). IEEE.

Murphy, E., Walsh, P.P., and Banerjee, A. (2021). *Framework for achieving the environmental sustainable development goals*. https://www.epa.ie/publications/research/climate-change/research-397-framework-for-achieving-the-environmental-sustainable-development-goals.php (accessed 8 June 2024).

Murugan, A., Nair, S.A.H., Preethi, A.A.P., et al. (2021). Diagnosis of skin cancer using machine learning techniques. *Microprocessors and Microsystems* 81: 103727.

Mutlu, E.Ç., Yousefi, N., and Ozmen Garibay, O. (2022). Contrastive counterfactual fairness in algorithmic decision-making. In *Proceedings of*

the 2022 AAAI/ACM Conference on AI, Ethics, and Society (AIES '22), pp. 499–507. NY, USA: Association for Computing Machinery.

Najibi, A. (2020). Racial discrimination in face recognition technology. *Science in the News*. https://sitn.hms.harvard.edu/flash/2020/racial-discrimination-in-face-recognition-technology/ (accessed 24 June 2024).

Nam, J., Cha, H., Ahn, S., et al. (2020). Learning from failure: de-biasing classifier from biased classifier. *Advances in Neural Information Processing Systems* 33 (pp. 20673–20684).

Nasarre, M., Strik, M., Ramirez, F.D., et al. (2022). Using a smartwatch electrocardiogram to detect abnormalities associated with sudden cardiac arrest in young adults. *EP Europace* 24 (3): 406–412.

Nguyen, A., Ngo, H., Hong, Y., et al. (2023). Ethical principles for artificial intelligence in education. *Education and Information Technologies* 28 (4): 4221–4241.

Nsengiyumva, J.B. and Valentino, R. (2020). Predicting landslide susceptibility and risks using GIS-based machine learning simulations, case of upper Nyabarongo catchment. *Geomatics, Natural Hazards and Risk* 11 (1): 1250–1277.

Office for Civil Rights (OCR) (2021). *HIPAA home*. HHS.gov. https://www.hhs.gov/hipaa/index.html (accessed 8 June 2024).

OECD (2024). *OECD legal instruments*. https://legalinstruments.oecd.org/en/instruments/OECD-LEGAL-0449 (accessed 27 June 2024).

Ohalete, N.C., Aderibigbe, A.O., Emmanuel Chigozie Ani, E.C., et al. (2023). AI-driven solutions in renewable energy: a review of data science applications in solar and wind energy optimization. *World Journal of Advanced Research and Reviews* 20 (3): 401–417.

OHCHR (2024). *Economic, social and cultural rights*. https://www.ohchr.org/en/human-rights/economic-social-cultural-rights (accessed 20 June 2024).

Omitaomu, O.A. and Niu, H. (2021). Artificial intelligence techniques in smart grid: a survey. *Smart Cities* 4 (2): 548–568.

Ozmen Garibay, O., Winslow, B., Andolina, S., et al. (2023). Six human-centered artificial intelligence grand challenges. *International Journal of Human–Computer Interaction* 39 (3): 391–437.

Padhee, S., Saha, T.K., Tetreault, J., et al. (2020). Clustering of social media messages for humanitarian aid response during crisis. *arXiv [cs.SI]*.

http://arxiv.org/abs/2007.11756.

Pai, S., Bontempi, D., Hadzic, I., et al. (2024). Foundation model for cancer imaging biomarkers. *Nature Machine Intelligence* 6 (3): 354–367.

Palaniappan, K., Lin, E.Y.T., and Vogel, S. (2024). Global regulatory frameworks for the use of artificial intelligence (AI) in the healthcare services sector. *Healthcare (Basel, Switzerland)* 12 (5): 562. https://doi.org/10.3390/healthcare12050562.

Patón-Romero, J.D., Vinuesa, R., Jaccheri, L., et al. (2022). State of gender equality in and by artificial intelligence. *IADIS-International Journal on Computer Science and Information Systems* 17 (2): 31–48. https://doi.org/10.33965/ijcsis_2022170203.

Pessach, D. and Shmueli, E. (2022). A review on fairness in machine learning. *ACM Computing Surveys* 55 (3): 1–44.

Pizzi, M., Romanoff, M., and Engelhardt, T. (2020). AI for humanitarian action: Human rights and ethics. *International Review of the Red Cross* 102 (913): 145–180.

Podvig, P. (2002). History and the current status of the Russian early-warning system. *Science and Global Security* 10 (1): 21–60.

Poli, M., Massaroli, S., Nguyen, E., et al. (2023). Hyena hierarchy: towards larger convolutional language models. *arXiv [cs.LG]*. http://arxiv.org/abs/2302.10866.

Rajabi, A. and Garibay, O.O. (2021). Towards fairness in AI: addressing bias in data using GANs. In *HCI International 2021—Late Breaking Papers: Multimodality, eXtended Reality, and Artificial Intelligence* (pp. 509–518). Springer International Publishing.

Rajabi, A. and Garibay, O.O. (2022). TabFairGAN: fair tabular data generation with generative adversarial networks. *Machine Learning and Knowledge Extraction* 4 (2): 488–501.

Ramaswamy, V.V., Kim, S.S.Y., and Russakovsky, O. (2021). Fair attribute classification through latent space DE-biasing. In *2021 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. IEEE. https://doi.org/10.1109/cvpr46437.2021.00918.

Rana, R., Kalia, A., Boora, A., et al. (2023). Artificial intelligence for surface water quality evaluation, monitoring and assessment. *Water* 15 (22): 3919.

Rashed-Al-Mahfuz, M., Haque, A., Azad, A., et al. (2021). Clinically applicable machine learning approaches to identify attributes of chronic kid-

ney disease (CKD) for use in low-cost diagnostic screening. *IEEE Journal of Translational Engineering in Health and Medicine* 9: 4900511.

Ray, P.P., Dash, D., and De, D. (2019). Edge computing for internet of things: a survey, e-healthcare case study and future direction. *Journal of Network and Computer Applications* 140: 1–22.

Remian, D. (2019). *Augmenting education: ethical considerations for incorporating artificial intelligence in education*. https://scholarworks.umb.edu/instruction_capstone/52/ (accessed 27 June 2024).

Ribeiro, M.T., Singh, S., and Guestrin, C. (2016). "Why should I trust you?": explaining the predictions of any classifier. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD '16)*, pp. 1135–1144. NY, USA: Association for Computing Machinery.

Rigby, M.J. (2019). Ethical dimensions of using artificial intelligence in health care. *AMA Journal of Ethics* 21 (2): 121–124.

Safdar, N.M., Banja, J.D., and Meltzer, C.C. (2020). Ethical considerations in artificial intelligence. *European Journal of Radiology* 122: 108768.

Saha, S., Pagnozzi, A., Bourgeat, P., et al. (2020). Predicting motor outcome in preterm infants from very early brain diffusion MRI using a deep learning convolutional neural network (CNN) model. *NeuroImage* 215: 116807.

Sarker, I.H. (2024). AI for critical infrastructure protection and resilience. In *AI-driven Cybersecurity and Threat Intelligence: Cyber Automation, Intelligent Decision-making and Explainability* (ed. I.H. Sarker), 153–172. Switzerland: Springer Nature.

Schneider, G. (2018). Automating drug discovery. *Nature Reviews Drug Discovery* 17 (2): 97–113.

SDGs (2016). *The 17 United Nations Sustainable Development Goals*. https://sdgs.un.org/goals (accessed 8 June 2024).

Sharifi, I., Goudarzi, S., and Khodabakhshi, M.B. (2019). A novel dynamical approach in continuous cuffless blood pressure estimation based on ECG and PPG signals. *Artificial Intelligence in Medicine* 97: 143–151.

Sharma, S., Karri, K., Ishwor Thapa, I., et al. (2016). Identifying enriched drug fragments as possible candidates for metabolic engineering. *BMC Medical Genomics* 9 (Suppl 2): 46.

Srivastava, D., Srivastava, S.K., Khan, S.B., et al. (2023). Early detection of lung nodules using a revolutionized deep learning model. *Diagnostics*

(Basel, Switzerland) 13 (22): 3485. https://doi.org/10.3390/diagnostics13223485.

Subbiah, V. (2023). The next generation of evidence-based medicine. *Nature Medicine* 29 (1): 49–58.

Sun, C., Feng, X.M., Fu, B., et al. (2023). Desertification vulnerability under accelerated dryland expansion. *Land Degradation & Development* 34 (7): 1991–2004.

Sun, W., Bocchini, P., and Davison, B.D. (2020). Applications of artificial intelligence for disaster management. *Natural Hazards* 103 (3): 2631–2689.

Tayebi, A., Yousefi, N., Yazdani-Jahromi, M., et al. (2022). UnbiasedDTI: mitigating real-world bias of drug-target interaction prediction by using deep ensemble-balanced learning. *Molecules* 27 (9): 2980. https://doi.org/10.3390/molecules27092980.

The White House (2022). *Blueprint for an AI bill of rights*. https://www.whitehouse.gov/ostp/ai-bill-of-rights/ (accessed 8 June 2024).

Tilburg Law Review (2003). Fundamental texts on European private law. *Tilburg Law Review* 11 (3): 708–708.

Tobore, T.O. (2019). On the need for the development of a cancer early detection, diagnostic, prognosis, and treatment response system. *Future Science OA* 6 (2): FSO439.

Tomašev, N., Cornebise, J., Hutter, F., et al. (2020). AI for social good: unlocking the opportunity for positive impact. *Nature Communications* 11 (1): 2468.

Tong, M.S., Zheng, H.Q., and Wan, G.C. (2019). Improvement of education method by using artificial intelligence technology. In *2019 IEEE International Conference on Engineering, Technology and Education (TALE)* (pp. 1–5). IEEE.

Udupa, S., Gagliardone, I., Deem, A., et al. (2020). *Hate speech, information disorder, and conflict*. https://doi.org/10.35650/md.2033.d.2020.

UNESCO (2019). Beijing consensus on artificial intelligence and education. In *International Conference on Artificial Intelligence and Education, Planning Education in the AI Era: Lead the leap*.

United Nations General Assembly (1949). *Universal declaration of human rights*. Department of State, United States of America.

United Nations High Commissioner for Refugees (UNHCR) (2019). *Project Jetson*. https://jetson.unhcr.org/ (accessed 14 June 2024).

United Nations Office for Disaster Risk Reduction (UNDRR) (2007). *Early warning system*. https://www.undrr.org/terminology/early-warning-system (accessed 10 June 2024).

United States Department of State (2023). *Artificial intelligence for accelerating progress on the sustainable development goals: addressing society's greatest challenges*. https://www.state.gov/artificial-intelligence-for-accelerating-progress-on-the-sustainable-development-goals-addressing-societys-greatest-challenges/ (accessed 20 June 2024).

US EPA (2016). *Global greenhouse gas overview*. https://www.epa.gov/ghgemissions/global-greenhouse-gas-overview (accessed 8 June 2024).

Vacarelu, F. and Aylett-Bullock, J. (2021). *Fusing AI into satellite image analysis to inform rapid response to floods*. UNITAR. https://unitar.org/about/news-stories/news/fusing-ai-satellite-image-analysis-inform-rapid-response-floods (accessed 21 June 2024).

Van Laere, E. and van Braak, J. (2017). Auditory support in linguistically diverse classrooms: factors related to bilingual text-to-speech use. *Journal of Computer Assisted Learning* 33 (1): 79–95.

van Leeuwen, K.G., de Rooij, M., Schalekamp, S., et al. (2022). How does artificial intelligence in radiology improve efficiency and health outcomes? *Pediatric Radiology* 52 (11): 2087–2093.

Verdant Robotics (2024). *Verdant robotics*. https://www.verdantrobotics.com/ (accessed 8 June 2024).

Verma, S. and Rubin, J. (2018). Fairness definitions explained. In *2018 IEEE/ACM International Workshop on Software Fairness (FairWare)* (pp. 1–7). IEEE.

Vochozka, M., Kliestik, T., Majerova, J., et al. (2018). Participating in a highly automated society: how artificial intelligence disrupts the job market. *Economics Management and Financial Markets* 13 (4): 57–62.

Wamsley, D. and Chin-Yee, B. (2021). COVID-19, digital health technology and the politics of the unprecedented. *Big Data & Society* 8 (1): 20539517211019441.

Wan, M., Zha, D., Liu, N., et al. (2023). In-processing modeling techniques for machine learning fairness: a survey. *ACM Transactions on Knowledge Discovery from Data* 17 (3): 1–27.

Wang, C. (2023). A review on 3D convolutional neural network. In *2023 IEEE 3rd International Conference on Power, Electronics and Computer Applications (ICPECA)* (pp. 1204–1208). IEEE.

Warren, R., Price, J., Forstenhäusler, N., et al. (2024). Risks associated with global warming of 1.5 to 4 °C above pre-industrial levels in human and natural systems in six countries. *Climatic Change* 177 (3): 48.

Wedajo, G.K. (2017). LiDAR DEM data for flood mapping and assessment; opportunities and challenges: a review. *Journal of Remote Sensing & GIS* 6 (4): 1000211. https://doi.org/10.4172/2469-4134.1000211.

West, D.M. and Allen, J.R. (2018). *How artificial intelligence is transforming the world*. Brookings. https://www.brookings.edu/articles/how-artificial-intelligence-is-transforming-the-world/ (accessed 20 June 2024).

Wever, M., Shah, M., and O'Leary, N. (2022). Designing early warning systems for detecting systemic risk: a case study and discussion. *Futures* 136: 102882.

What-If Tool (2024). *What-If tool*. https://pair-code.github.io/what-if-tool/ (accessed 8 June 2024).

Winslow, B.D., Chadderdon, G.L., Dechmerowski, S.J., et al. (2016). Development and clinical evaluation of an mHealth application for stress management. *Frontiers in Psychiatry* 7: 130.

Winslow, B.D., Kwasinski, R., Whirlow, K.W., et al. (2022). Automatic detection of pain using machine learning. *Frontiers in Pain Research (Lausanne, Switzerland)* 3: 1044518.

World Food Programme (WFP) (2024). *HungerMap LIVE*. https://hungermap.wfp.org/ (accessed 14 June 2024).

Wynants, L., Calster, B.V., Collins, G., et al. (2020). Prediction models for diagnosis and prognosis of COVID-19: systematic review and critical appraisal. *BMJ* 369: m1328.

Yale School of Medicine (2021). *Hard choices: AI in health care*. https://medicine.yale.edu/news/yale-medicine-magazine/article/hard-choices-ai-in-health-care/ (accessed 20 June 2024).

Xia, Y., Sun, M., Huang, H., et al. (2024). Drug repurposing for cancer therapy. *Signal Transduction and Targeted Therapy* 9 (1): 92.

Zhang, S. and Metaxas, D. (2024). On the challenges and perspectives of foundation models for medical image analysis. *Medical Image Analysis* 91: 102996.

Zhou, H.-Y., Guo, J., Zhang, Y., et al. (2023). nnFormer: volumetric medical image segmentation via a 3D transformer. *IEEE Transactions on Image Processing* 32: 4036–4045.

Zhou, Z., Siddiquee, M.M.R., Tajbakhsh, N., et al. (2018). UNet++: a nested U-Net architecture for medical image segmentation. In *Deep Learning in Medical Image Analysis and Multimodal Learning for Clinical Decision Support* (pp. 3–11). Springer International Publishing.